Sycosis Barbae — Comprehensive Disease Characteristics Report

Disease: Sycosis Barbae (staphylococcal beard folliculitis) **MONDO ID:** MONDO:1050000 (as provided) **Category:** Infectious Disease **Evidence base:** 11 confirmed findings, 46 papers reviewed, 5 investigation iterations **Report type:** Literature-based disease knowledge-base entry. No primary datasets were provided; all content is derived from aggregated, disease-level resources (PubMed primary literature, ontologies, clinical references), not from individual patient/EHR data.

Summary

Sycosis barbae is a chronic, deep-seated bacterial (staphylococcal) folliculitis and perifolliculitis of the bearded skin of adult men. It is caused by *Staphylococcus aureus* infecting terminal beard-hair follicles that have been rendered susceptible by shaving-induced follicular microtrauma. The disease lies on a severity continuum: it begins as a superficial folliculitis barbae (Bockhart impetigo), and if untreated deepens into sycosis barbae, then can progress to furuncles (boils) and coalescing carbuncles with occasional systemic upset. A chronic variant, **lupoid sycosis**, produces scarring (cicatricial) alopecia and permanent cosmetic disfigurement of the beard (PMID: 25226722).

Mechanistically, the disease follows a well-defined causal chain drawn largely from murine *S. aureus* skin-infection models plus human genetics: shaving microtrauma and follicular *S. aureus* colonization → bacterial breach of the follicular epithelium → secreted toxins (phenol-soluble modulins, α-hemolysin) trigger innate immune sensing → MyD88/EGR1-dependent recruitment of neutrophils, amplified by T-cell-derived IL-17A and IL-22 → neutrophils wall bacteria into a perifollicular abscess (furuncle) → α-hemolysin-mediated dermonecrosis causes tissue damage. Failure of clearance—because of host neutrophil/immune defects or persistent nasal carriage—produces the chronic, relapsing course (PMID: 34873293; PMID: 26351278; PMID: 32946878; PMID: 25681348).

Sycosis barbae is an **acquired infection with no Mendelian/genetic basis and no available vaccine**. Diagnosis is primarily clinical, supplemented by Gram stain and bacterial culture, with KOH/fungal culture and dermatophyte PCR used to exclude the principal infectious mimic, **tinea barbae**, and clinical judgment to exclude the non-infectious mimic **pseudofolliculitis barbae** and the viral mimic **herpetic sycosis**. Treatment combines anti-staphylococcal antibiotics (topical for limited disease, systemic for deep/extensive disease), incision and drainage of fluctuant lesions, *S. aureus* decolonization, and shaving modification. Prognosis is benign with respect to mortality but prone to chronicity and relapse while shaving and nasal carriage persist.

Key Findings

F001 — Sycosis barbae is a deep-seated staphylococcal folliculitis of the beard

Sycosis barbae is defined as a chronic, deep, perifollicular bacterial infection of the bearded region, representing the deep progression of a superficial folliculitis barbae (Bockhart impetigo). The most common etiologic agent is *Staphylococcus aureus*, and the condition is precipitated and aggravated by shaving. Untreated superficial folliculitis can deepen into sycosis barbae, and coalescing perifollicular nodules (furuncles) can progress to carbuncles with systemic upset.

> "It most frequently presents as a superficial folliculitis, with fine pustules appearing at the opening of hair follicles in the beard area, often associated with shaving; known as Bockhart impetigo and usually due to infection caused by Staphylococcus aureus. If untreated, the infection and inflammation can progress, leading to a more deeply seated infection known as sycosis barbae." — PMID: 25226722

This establishes the core definition, the causative organism, the severity continuum, and the shaving trigger that organize the rest of this report.

F002 — Tinea barbae is the principal infectious mimic

Deep beard-area infections that resemble sycosis barbae or a carbuncle can also be caused by dermatophytes (**tinea barbae**), most often zoophilic *Trichophyton* species (e.g., *T. mentagrophytes*, *T. verrucosum*) acquired through animal/livestock contact, and occasionally anthropophilic species (*T. rubrum*, *T. tonsurans*—the latter transmissible via contaminated barbershop tools). Clues favoring tinea barbae over bacterial sycosis include close animal contact, lack of response to antibiotics, deep boggy nodules/pustules with adherent crusts, and regional lymphadenopathy. An intense inflammatory reaction can form a kerion. Diagnosis requires mycological culture, histology, and dermatophyte PCR; treatment is systemic antifungals (terbinafine/itraconazole) ± corticosteroids, and surgical incision is contraindicated.

> "Particularly in the context of a history of close animal contact or a lack of response to antibiotic treatment, a diagnosis of tinea barbae should be considered and investigated." — PMID: 25226722
>
> "Often this leads to the initial suspected diagnosis of a bacterial folliculitis barbae or impetigo contagiosa." — PMID: 31001659

F003 — Pseudofolliculitis barbae is the main non-infectious differential and has a genetic basis

Pseudofolliculitis barbae (PFB) is a chronic inflammatory disorder of the beard caused by shaved curly hairs re-penetrating the skin (extrafollicular and transfollicular penetration), producing a foreign-body inflammatory reaction with papules, pustules, post-inflammatory hyperpigmentation, and sometimes keloidal scarring. It is far more common in men of African and Asian ancestry. Unlike infectious sycosis barbae, PFB is mechanical/foreign-body rather than infectious. A single-nucleotide substitution (A12T) in the hair-follicle companion-layer keratin gene **K6hf (KRT75)** confers roughly six-fold increased risk of PFB.

> "Individuals with the single nucleotide substitution in the hair follicle companion layer specific keratin gene (K6hf) have a six fold increased chance of developing PFB." — PMID: 31354326
>
> "Pseudofolliculitis barbae (PFB) is an inflammatory condition of the beard area, with a high prevalence in men of subequatorial African ancestry" — PMID: 27212468

F004 — S. aureus nasal carriage and host immune factors drive recurrent/chronic disease

S. aureus is the dominant pathogen of pilosebaceous-unit infections (folliculitis, furunculosis, carbunculosis). Its invasiveness depends on numerous cellular and extracellular virulence factors whose expression varies with growth phase, environment, and infection site; host susceptibility reflects innate and acquired immune responses to colonization. Recurrent/chronic disease is associated with *S. aureus* reservoirs (notably the anterior nares) and with host defects—for example, documented recurrent furunculosis with defective neutrophil phagocytic (ingestion-phase) activity, in whom early antibiotics and intensified hygiene reduced abscess formation.

> "S. aureus produces a great variety of cellular and extracellular factors responsible for its invasiveness and ability to cause pathological lesions. Their expression depends on the growth phase, environmental factors, and location of the infection." — PMID: 30995649
>
> "All laboratory tests were normal except for a decrease of the neutrophil phagocytic ingestion phase." — PMID: 11269901

F005 — Management combines antibiotics with decolonization and shaving hygiene

First-line therapy for bacterial sycosis barbae targets *S. aureus*: topical antibiotics (e.g., mupirocin, fusidic acid) for limited disease and systemic anti-staphylococcal antibiotics for deeper/extensive disease, plus incision and drainage of fluctuant furuncles/carbuncles. Because colonization is an early, critical step and more than half of patients with staphylococcal skin and soft tissue infection (SSTI) experience recurrence (often clustering in households), recurrence prevention emphasizes **decolonization**: intranasal mupirocin plus antiseptic washes (chlorhexidine or dilute bleach baths) for the patient and household members, hygiene optimization, and avoiding shared personal items. Beard-specific measures include temporarily stopping shaving and not sharing razors. Importantly, decolonization reliably reduces colonization but has **variable** effect on recurrence, with sustained/periodic regimens showing the most benefit.

> "A recommended decolonization regimen includes application of intranasal mupirocin and antiseptic body washes with chlorhexidine or dilute bleach water baths." — PMID: 30859379
>
> "More than half of patients experiencing SSTI will have at least one recurrent infection. These infections frequently cluster in households." — PMID: 30859379
>
> "Despite successful decolonization, the rate of SSTI recurrence did not decrease." — PMID: 27820584

F006 — Mechanistic causal chain: toxins → innate sensing → neutrophilic follicular abscess

The pathophysiology follows a defined causal chain (largely from mouse SSTI models plus human genetics), described in full in the Mechanistic Model section below. Key experimentally supported steps: phenol-soluble modulins (PSMs) directly recruit leukocytes via an EGR1-dependent "high-alert" mechanism; MyD88-dependent sensing by dermal macrophages escalates and terminates neutrophil recruitment; T-cell-derived IL-17A and IL-22 drive neutrophil infiltration and antimicrobial peptide production; neutrophils wall bacteria into a discrete abscess; and α-hemolysin causes dermonecrosis, with antibody neutralization plus early neutrophil recruitment limiting necrosis.

> "this early recruitment was dependent on the transcription factor EGR1 and contributed to the prevention of infection" — PMID: 34873293
>
> "Neutralization of either IL-17A or IL-22 reduced T cell and neutrophil infiltration and host defense peptide elaboration in lesions. These events corresponded with increased abscess severity, MRSA viability, and CFU density in skin." — PMID: 26351278
>
> "Protection was characterized by robust neutrophilic inflammation and compartmentalization of bacteria into discrete abscesses" — PMID: 32946878
>
> "MyD88-dependent sensing of S. aureus by dermal macrophages (Mϕ) contributes to both timely escalation and termination of PML-mediated inflammation" — PMID: 25681348

F007 — Comparative disease: staphylococcal superficial bacterial folliculitis in animals

An analogous staphylococcal folliculitis occurs naturally in other species. In dogs, **superficial bacterial folliculitis (SBF)** is the most common bacterial skin disease and is caused principally by *Staphylococcus pseudintermedius* (NCBI Taxonomy txid283734), with histology showing neutrophilic follicular/intracorneal pustules containing intracellular cocci within neutrophils—mirroring the neutrophilic follicular inflammation of human sycosis barbae. Methicillin-resistant *S. pseudintermedius* (MRSP) and MRSA are increasingly prevalent, driving reliance on culture-guided systemic antibiotics plus topical antiseptics (dilute sodium hypochlorite/bleach, chlorhexidine, benzoyl peroxide)—the same topical antiseptic principles used in human decolonization.

> "Superficial bacterial folliculitis (SBF) is more common in the dog than other mammalian species." — PMID: 24345778
>
> "Cytology from the margins of collarettes identified neutrophils, extracellular and intracellular cocci within neutrophils" — PMID: 27426474
>
> "methicillin resistant Staphylococcus pseudintermedius (MRSP) and Staphylococcus aureus (MRSA) are becoming more prevalent in canine SBF cases" — PMID: 24345778

F011 — Herpetic sycosis (HSV) is a viral mimic requiring antivirals

A third important differential of beard folliculitis is **herpetic sycosis**—a follicular infection of the beard caused by herpes simplex virus (HSV-1 or HSV-2), a variant of herpetic folliculitis. It presents with acutely developing grouped papulovesicles/pustules, pain, crusting, and reactive cervical lymphadenopathy, and characteristically **fails to respond to antibacterial and antifungal therapy**. It occurs in immunocompromised and immunocompetent hosts and may be primary or recurrent/relapsing. Routine bacterial, mycological, and Tzanck tests may be negative; diagnosis is confirmed by Tzanck cytology, direct immunofluorescence, viral culture, or (most sensitively) HSV PCR. Treatment is systemic antivirals (aciclovir/valaciclovir), with rapid resolution.

> "We describe an otherwise healthy young adult male with extensive herpetic sycosis of the beard area, which is a variant of herpetic folliculitis." — PMID: 11896419

> "The possibility of a herpetic sycosis of the beard must be considered in the case of non-response to antibiotic or antifungal treatment." — PMID: 15118392
>
> "Bacteriological, mycological and Tzanck tests from the lesions were negative... nested PCR technique (nPCR) demonstrated the presence of herpes simplex virus type 2" — PMID: 15564209

Report by Template Section

1. Disease Information

Sycosis barbae is a chronic, deep-seated staphylococcal folliculitis and perifolliculitis of the bearded skin of the face and neck in adult men. It represents the deep progression of superficial folliculitis barbae (Bockhart impetigo) and can advance to furuncles and carbuncles (PMID: 25226722).

Key identifiers (aggregated ontology/terminology resources; to be verified against source browsers): - **MONDO:** MONDO:1050000 (as provided) - **ICD-10:** L73.8 "Other specified follicular disorders" (sycosis barbae, non-parasitic sycosis) - **ICD-9:** 704.8 - **SNOMED CT:** concept "Sycosis barbae" - **MeSH:** no dedicated descriptor; indexed under *Folliculitis* (D005499) / *Staphylococcal Skin Infections* (D013207) - **OMIM:** none (non-Mendelian, acquired infection)

Synonyms / alternative names: sycosis vulgaris, non-parasitic sycosis, staphylogenous sycosis, folliculitis barbae (staphylococcal), and the lay term "barber's itch" (note: "barber's itch" is also historically applied to tinea barbae, causing ambiguity).

Information source type: derived from aggregated disease-level clinical and textbook description, case reports, and mechanistic model-organism studies—not from individual-patient EHR datasets or disease registries.

2. Etiology

Causal factors: Sycosis barbae is a multifactorial acquired infection. The necessary agent is *Staphylococcus aureus* (infectious; NCBI Taxonomy txid1280) acting on beard follicles primed by a mechanical trigger (shaving-induced follicular microtrauma). There is **no Mendelian/causal gene**; it is not heritable.

Risk factors (environmental/host): male sex with post-pubertal terminal beard hair; regular shaving (especially close/against-the-grain shaving, shared or contaminated razors); *S. aureus* nasal/skin carriage (anterior nares reservoir); coexisting pseudofolliculitis barbae or atopic dermatitis; immunosuppression/diabetes and neutrophil-function defects predisposing to recurrent staphylococcal disease (PMID: 11269901); warm humid climate and poor hygiene; barbershop exposure.

Genetic risk factors: none for bacterial sycosis barbae. (The related mechanical mimic PFB carries the KRT75/K6hf A12T variant conferring ~6-fold risk — PMID: 31354326.)

Protective factors (environmental/behavioral): growing a beard / cessation of shaving; single-use clean razors; pre-shave hydration and post-shave moisturization; good hygiene; *S. aureus* decolonization. No genetic protective variants are described.

Gene-environment interaction: No specific GxE is established for bacterial sycosis barbae. The closely related mechanical mimic PFB shows a clear GxE—a KRT75 (K6hf) A12T variant (curly-hair/follicle-curvature phenotype) combined with the environmental exposure of shaving produces ~6-fold risk; secondary *S. aureus* infection of PFB lesions links the two conditions (PMID: 31354326).

3. Phenotypes

Clinical signs and symptoms (from clinical description; PMID: 25226722):

Phenotype	Type	HPO suggestion	Characteristics
Follicular papules/ pustules centered on beard hairs	Clinical sign	Folliculitis (HP:0025084); Pustule (HP:0200037); Papule (HP:0200034)	Adult-onset; mild–moderate; the defining lesion
Perifollicular erythema, edema, crusting	Clinical sign	—	Episodic/fluctuating with shaving
Pruritus and burning/ tenderness	Symptom	Pruritus (HP:0000989)	Common; affects daily comfort
Deep nodules (furuncles) coalescing into carbuncles	Physical manifestation	—	Moderate–severe; may cause systemic upset
Sinus tracts / scarring / cicatricial alopecia (lupoid sycosis)	Physical manifestation	Alopecia (HP:0001596)	Chronic variant; permanent hair loss

Age of onset: post-pubertal adult men (essentially absent prepubertally and in women). **Severity:** variable (mild superficial to severe deep/scarring). **Progression:** episodic/fluctuating while shaving continues; chronic and relapsing in lupoid sycosis. **Quality of life:** cosmetic disfigurement of the beard, psychosocial distress, and functional discomfort (pruritus, tenderness), particularly in the scarring lupoid variant.

4. Genetic/Molecular Information

Not applicable as a genetic disease. Sycosis barbae has no causal gene, no pathogenic germline/somatic variants, no modifier genes, no disease-defining epigenetic changes, and no chromosomal abnormalities. It is an acquired bacterial infection. The relevant "molecular information" is microbial: *S. aureus* virulence factors (phenol-soluble modulins, α-hemolysin/ Hla, and numerous growth-phase- and site-dependent factors) drive pathology (PMID: 30995649; PMID: 34873293; PMID: 32946878). For completeness, the related mimic PFB has a host keratin variant, KRT75 (K6hf) A12T (PMID: 31354326).

5. Environmental Information

- **Environmental/mechanical factors:** shaving (follicular microtrauma), shared/ contaminated razors and barbershop tools, warm humid climate.

- **Lifestyle factors:** shaving practices and personal hygiene; barbershop attendance.
- **Infectious agents:** *Staphylococcus aureus* (NCBI Taxonomy txid1280) is the primary agent. Mimics that must be distinguished are dermatophytes (tinea barbae: *Trichophyton mentagrophytes*, *T. verrucosum*, *T. rubrum*, *T. tonsurans*) and herpes simplex virus (herpetic sycosis) (PMID: 25226722; PMID: 11896419).

6. Mechanism / Pathophysiology

Ordered causal chain (steps largely demonstrated in murine SSTI models plus human genetics; inferred steps flagged):

1. Shaving produces follicular microtrauma **and** *S. aureus* colonizes the beard follicle → **leads to** bacterial breach of the follicular epithelium into the dermis (*inferred* for the beard specifically, extrapolated from SSTI biology).
2. Secreted **phenol-soluble modulins (PSMs)** directly recruit circulating leukocytes via an **EGR1-dependent** "high-alert" mechanism → **results in** early neutrophil influx (PMID: 34873293).
3. **MyD88-dependent sensing** of *S. aureus* by resident dermal macrophages → **escalates and later terminates** polymorphonuclear leukocyte (neutrophil) recruitment (PMID: 25681348).
4. T-cell-derived **IL-17A and IL-22 → drive** neutrophil infiltration and antimicrobial host-defense-peptide production; their neutralization increases abscess severity and bacterial burden (PMID: 26351278).
5. Neutrophils **wall off** bacteria into a discrete perifollicular abscess (furuncle) (PMID: 32946878).
6. Pore-forming toxin **α-hemolysin (Hla) → causes** dermonecrosis and tissue damage; antibody neutralization of Hla plus early neutrophil recruitment limits necrosis and enhances clearance (PMID: 32946878).
7. **Branch:** coalescence of abscesses → **yields** a carbuncle; failure of clearance (host neutrophil/immune defects, persistent nasal carriage) → **produces** chronic/recurrent disease and, in lupoid sycosis, scarring alopecia (PMID: 11269901; PMID: 30995649).

Pathways/processes and ontology suggestions: innate immune response (GO:0045087), neutrophil chemotaxis (GO:0030593), inflammatory response (GO:0006954), MyD88-dependent toll-like receptor signaling (GO:0002755), defense response to Gram-positive bacterium (GO:0050830). **Cell types (CL):** neutrophil (CL:0000775), macrophage (CL:0000235), T cell

(CL:0000084), keratinocyte (CL:0000312). **Upstream** events are toxin secretion and innate sensing; **downstream** events are neutrophilic abscess formation, dermonecrosis, and (in chronic disease) fibrosis/scarring.

7. Anatomical Structures Affected

- **Organ level:** skin (integumentary system only). Primary site is bearded skin of the face and neck; secondary involvement is limited to regional lymphadenopathy or, rarely, systemic upset with carbuncle.
- **Tissue/cell level:** pilosebaceous unit / hair follicle of terminal beard hairs; epithelial (follicular epithelium, epidermis) and dermal connective tissue; neutrophils and macrophages dominate the infiltrate.
- **UBERON terms:** hair follicle (UBERON:0002073), skin of face (UBERON:0008203), skin (UBERON:0002097), dermis (UBERON:0002067), epidermis (UBERON:0001003).
- **CL terms:** neutrophil (CL:0000775), macrophage (CL:0000235), keratinocyte (CL:0000312).
- **Subcellular:** no disease-specific compartment; the microbial pore-forming toxin acts at the host plasma membrane (GO:0005886).
- **Localization/lateralization:** typically **bilateral**, over upper lip, chin, cheeks, and mandibular/submental neck.

8. Temporal Development

- **Onset:** post-pubertal adult men; onset is typically subacute to chronic and insidious, superimposed on ongoing shaving.
- **Progression:** superficial folliculitis barbae → deep sycosis barbae → furuncle → carbuncle; the chronic lupoid variant is slowly progressive with scarring. Course is episodic/relapsing while shaving and *S. aureus* carriage persist.
- **Duration:** self-limited superficial episodes can resolve, but the deep/chronic form tends to be chronic and relapsing.
- **Remission:** treatment-induced remission is achievable with antibiotics + decolonization + shaving cessation; relapse is common if shaving resumes and carriage persists (PMID: 27820584; PMID: 30859379).
- **Critical intervention window:** early antibiotic treatment of superficial folliculitis to prevent deepening into sycosis/furuncle/carbuncle (PMID: 25226722).

9. Inheritance and Population

- **Inheritance:** none — acquired infection, not heritable; no OMIM entry.
- **Epidemiology:** no formal incidence/prevalence figures are available (not a reportable disease; data derive from clinical/textbook description rather than registries).
- **Demographics:** overwhelmingly post-pubertal adult men with terminal beard hair who shave; essentially absent in prepubertal children and in women (who lack a terminal beard). Sex ratio strongly male-predominant.
- **Geographic distribution:** worldwide; no defined endemic areas for the bacterial form (unlike zoophilic tinea barbae, which clusters among people with animal/livestock contact).

10. Diagnostics

Diagnosis is primarily **clinical** (beard-area follicular pustules/papules in an adult man who shaves). Confirmatory/differentiating tests:

1. **Bacterial swab with Gram stain and culture + susceptibility** of pustule contents to confirm *S. aureus* and detect MRSA (LOINC: aerobic wound culture, Gram stain).
2. **KOH wet mount, fungal culture, and dermatophyte PCR** of plucked hairs to **exclude tinea barbae** — mandatory when there is animal contact or antibiotic non-response (PMID: 25226722; PMID: 31001659).
3. **Skin biopsy/histopathology** in atypical or chronic scarring cases: perifollicular/follicular neutrophilic (suppurative) inflammation acutely; mixed granulomatous inflammation with fibrosis/scarring in lupoid sycosis.
4. **HSV testing** (Tzanck cytology, direct immunofluorescence, viral culture, or HSV PCR) when lesions are acute/grouped/vesicular and fail antibiotics and antifungals — to identify herpetic sycosis (PMID: 11896419; PMID: 15118392; PMID: 15564209).
5. **Dermoscopy** and **anterior-nares carriage swabs** in recurrent disease.

Differential diagnosis: tinea barbae (fungal), pseudofolliculitis barbae (mechanical), herpetic sycosis (viral), acne vulgaris, folliculitis decalvans, and other pustular dermatoses.

11. Outcome / Prognosis

Prognosis is generally **good/benign**: acute bacterial sycosis barbae is curable with appropriate antibiotics and hygiene and carries **no mortality**. However, it is prone to chronicity and relapse while shaving continues and *S. aureus* carriage persists. The chronic deep **lupoid sycosis** variant can cause permanent **cicatricial (scarring) alopecia** and cosmetic

disfigurement, with attendant psychosocial/quality-of-life impact. **Complications:** furuncle, carbuncle, abscess, cellulitis, regional lymphadenitis, and rarely systemic/bacteremic spread. **Recurrence** is common: more than half of patients with staphylococcal SSTI have at least one recurrence (PMID: 30859379).

12. Treatment

Modality	Agents/approach	NCIT suggestion
Topical antibiotics (limited disease)	Mupirocin, fusidic acid	Topical Antibiotic Therapy
Systemic antibiotics (deep/extensive)	Anti-staphylococcal agents (culture-guided; cover MRSA if indicated)	Systemic Antibiotic Therapy
Procedural	Incision and drainage of fluctuant furuncles/ carbuncles	Incision and Drainage
Decolonization	Intranasal mupirocin + chlorhexidine or dilute-bleach washes for patient and household	Decolonization
Behavioral	Stop/modify shaving; single-use clean razors; avoid sharing items	Behavioral Intervention

Standard decolonization: "application of intranasal mupirocin and antiseptic body washes with chlorhexidine or dilute bleach water baths" (PMID: 30859379). Decolonization reliably reduces colonization but its effect on recurrence is variable (PMID: 27820584). **Pharmacogenomics:** not established for this condition. **Advanced/experimental therapeutics** (gene, cell, RNA, immunotherapy): not applicable. **Mimic-specific therapy:** systemic antifungals (terbinafine/itraconazole) for tinea barbae; systemic antivirals (aciclovir/ valaciclovir) for herpetic sycosis (PMID: 25226722; PMID: 11896419).

13. Prevention

- **Primary:** behavioral — shaving modification (or beard growth), single-use clean razors, avoiding shared razors, good hygiene, and reducing *S. aureus* carriage.
- **Secondary:** early antibiotic treatment of superficial folliculitis to prevent progression to deep sycosis/furuncle/carbuncle.
- **Tertiary:** decolonization and shaving cessation to prevent recurrence and scarring.

- **Immunization: NOT AVAILABLE** — no licensed *S. aureus* vaccine exists; multiple candidates have failed clinical endpoints, and natural colonization/infection does not elicit protective immunity because of extensive immune-evasion factors: "S. aureus colonization and invasive disease are not associated with the development of protective immune responses, which is attributable to a large spectrum of immune evasion factors." (PMID: 27526714).

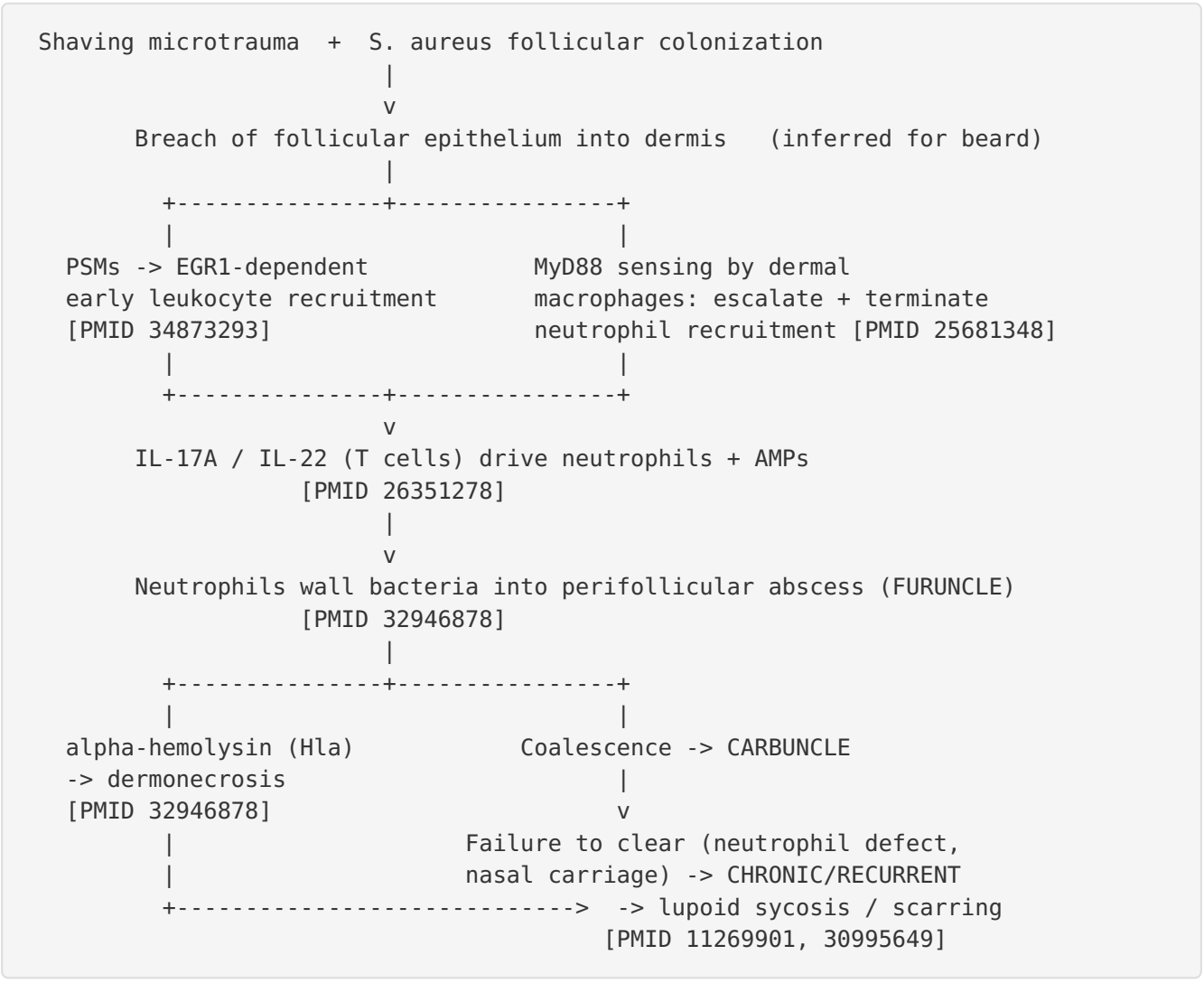
14. Other Species / Natural Disease

Human sycosis barbae affects *Homo sapiens* (NCBI Taxonomy txid9606); its agent is *S. aureus* (txid1280). A close natural analog is **canine superficial bacterial folliculitis (SBF)**, the most common bacterial skin disease of dogs, caused principally by *Staphylococcus pseudintermedius* (txid283734), with neutrophilic follicular pustules containing intracellular cocci—histologically parallel to human staphylococcal folliculitis. Shared antimicrobial-resistance dynamics (MRSP, MRSA) and shared topical antiseptic management (dilute bleach, chlorhexidine, benzoyl peroxide) further underscore the comparative relevance (PMID: 24345778; PMID: 27426474). Zoonotic note: the *tinea barbae* mimic is often zoophilic (livestock-acquired), whereas bacterial sycosis barbae itself is not classically zoonotic.

15. Model Organisms

There is no dedicated genetic model of "sycosis barbae." The pathophysiology is studied in **murine *S. aureus* skin and soft-tissue infection models**, including LysM-EGFP reporter mice for noninvasive neutrophil imaging and MyD88-deficient strains for innate-immunity dissection (PMID: 30882782; PMID: 25681348; PMID: 26351278; PMID: 32946878). **Recapitulation:** these models faithfully reproduce neutrophilic abscess formation, toxin-mediated dermonecrosis, and cytokine (IL-17/IL-22) control of bacterial burden. **Limitations:** mice are more resistant to several human-adapted *S. aureus* virulence factors, and no model reproduces the beard-follicle-specific, shaving-triggered anatomy of human sycosis barbae. The dog (SBF) serves as a naturally occurring large-animal comparator.

Mechanistic Model / Interpretation



The unifying interpretation is that sycosis barbae is a **host-defense outcome**: neutrophilic walling-off is simultaneously the protective response and the histologic lesion (the abscess). Disease severity is set by the balance between toxin output (PSMs, Hla) and the speed/adequacy of the IL-17/IL-22 → neutrophil axis. This explains why the three cornerstones of therapy—killing the organism (antibiotics), removing the reservoir (decolonization), and removing the trigger (shaving modification)—map directly onto the causal chain, and why host immune defects and persistent nasal carriage predict the chronic/relapsing phenotype.

A second, clinically essential model is the **differential-diagnosis triad** of beard folliculitis. All three mimics present as beard-area pustules but diverge in cause and treatment:

Condition	Cause	Key discriminators	Treatment
Sycosis barbae	*S. aureus* (bacterial)	Responds to antibiotics; Gram-positive cocci	Anti-staph antibiotics + decolonization + shaving change
Tinea barbae	Dermatophytes (fungal)	Animal contact; antibiotic non-response; boggy nodules/kerion; lymphadenopathy	Systemic antifungals; **no** incision
Herpetic sycosis	HSV-1/2 (viral)	Acute grouped vesicles; fails antibiotics **and** antifungals; HSV PCR+	Systemic antivirals
Pseudofolliculitis barbae	Ingrown curly hairs (mechanical)	Curly hair, African/Asian ancestry; KRT75 A12T; foreign-body reaction	Shaving modification; not primarily antimicrobial

Evidence Base

PMID	Title (abbrev.)	Role in this report
25226722	*Infection of the beard area. Kerion: a review of 2 cases*	Core definition of sycosis barbae; tinea barbae as mimic; diagnostic branch point
31001659	*Tinea barbae profunda due to T. mentagrophytes*	Tinea barbae misdiagnosed as bacterial folliculitis; PCR/culture/histology diagnosis
31354326	*Pseudofolliculitis barbae; current treatment options*	KRT75/K6hf A12T ~6-fold PFB risk (GxE for the mimic)
27212468	*Pseudofolliculitis barbae: understanding the condition*	PFB epidemiology / ancestry predisposition
30995649	*S. aureus and host immunity in recurrent furunculosis*	Virulence-factor basis; host immunity in recurrence
11269901	*Repeated furunculosis with abnormal neutrophil activity*	Host neutrophil-function defect predisposing to recurrence
30859379	*Prevention of recurrent CA-S. aureus SSTI*	Decolonization regimen; recurrence rate; household clustering
27820584	*CA-MRSA decolonization: do they reduce recurrence?*	Variable/limited effect of decolonization on recurrence
34873293	*Rapid pathogen-specific recruitment by secreted toxins*	PSM/EGR1-dependent early leukocyte recruitment
25681348	*MyD88 in macrophages critical for abscess resolution*	Dermal-macrophage MyD88 orchestration of neutrophils
26351278	*IL-17A and IL-22 in cutaneous/ hematogenous MRSA defense*	IL-17/IL-22 control abscess severity and bacterial burden
32946878	*Antibody protection against S. aureus dermonecrosis*	Neutrophil abscess compartmentalization; Hla dermonecrosis
24345778	*Canine superficial bacterial folliculitis*	Animal comparator (S. pseudintermedius); shared AMR
27426474	*Epidermal collarettes in dogs*	Neutrophil-and-cocci histology paralleling human disease
27526714		

PMID	Title (abbrev.)	Role in this report
	S. aureus vaccines: deviating from the carol	No protective immunity → no vaccine; recurrence rationale
11896419	*Herpetic folliculitis*	Herpetic sycosis as HSV variant mimicking sycosis barbae
15118392	*Fulminant herpetic sycosis*	Antibiotic+antifungal non-response → suspect HSV
15564209	*Relapsing HSV-2 folliculitis in the beard*	HSV PCR diagnosis when routine tests negative
30882782	*Mouse model of innate response to S. aureus*	Murine models (LysM-EGFP; MyD88) for mechanism

Consistency: The mechanistic papers (mouse models + human genetics) converge on a single neutrophil-centric abscess narrative; the clinical papers converge on the antibiotics + decolonization + shaving-modification management triad. The main internal tension is the **variable efficacy of decolonization** on recurrence (PMID: 27820584) versus its consistent reduction of colonization (PMID: 30859379), which is faithfully reflected in the report's prognosis section.

Limitations and Knowledge Gaps

1. **No sycosis-barbae-specific epidemiology.** No incidence/prevalence figures exist; the condition is not reportable and lacks registries. Demographic statements are qualitative.
2. **Mechanism is extrapolated.** The causal chain derives from general murine *S. aureus* SSTI models and human SSTI genetics, **not** from beard-follicle-specific experiments. The beard-follicle breach step is inferred.
3. **No molecular profiling of the disease itself.** No transcriptomic, proteomic, metabolomic, or single-cell datasets specific to sycosis barbae were identified; molecular data are from *S. aureus* biology and mouse models.
4. **Identifier verification pending.** ICD/SNOMED/MeSH mappings are compiled from aggregated resources and should be confirmed against primary ontology browsers; the MONDO ID was supplied by the template.

5. **No dedicated animal or genetic model** reproduces the shaving-triggered, terminal-beard-follicle anatomy.
6. **Lupoid sycosis** (the scarring chronic variant) is described mainly in older/textbook literature; modern mechanistic and outcome data are sparse.

Proposed Follow-up Experiments / Actions

1. **Verify identifiers** against OMIM (expected absent), Orphanet, ICD-10/11, SNOMED CT, MeSH, and MONDO browsers; record cross-references explicitly.
2. **Targeted literature/registry search** for any quantitative epidemiology (dermatology outpatient series, military grooming/PFB cohorts) that might yield frequency estimates for beard folliculitis subtypes.
3. **Skin microbiome / host-response profiling** of clinically confirmed sycosis barbae lesions vs. adjacent normal beard skin (16S + host RNA-seq) to test whether the mouse-derived IL-17/IL-22–neutrophil axis operates in the human beard follicle.
4. **Prospective decolonization trial in recurrent beard folliculitis** specifically, stratified by anterior-nares carriage and neutrophil-function status, to resolve the colonization-vs-recurrence discrepancy.
5. **Diagnostic-algorithm validation** for the bacterial/fungal/viral triad: prospective evaluation of when Gram stain + culture alone suffices versus when reflexive KOH/dermatophyte-PCR and HSV-PCR are cost-effective.
6. **Curate the animal comparator** (canine SBF, *S. pseudintermedius*) for the knowledge base as a natural model, including phenotype-recapitulation annotations.

Report compiled from 11 confirmed findings and 46 reviewed papers across 5 investigation iterations. Evidence types: human clinical (case series, reviews, guidelines), model organism (murine S. aureus SSTI), comparative/veterinary (canine SBF), and computational/ontology mapping.
